

Pre-Crash Scenario Analysis and Generation for Integrated Safety System Virtual Testing with Real Accident Data

Author, co-author (Do NOT enter this information. It will be pulled from participant tab in MyTechZone)

Affiliation (Do NOT enter this information. It will be pulled from participant tab in MyTechZone)

Abstract

Integrated active and passive safety protection systems have made substantial contributions to reducing traffic accidents and mitigating human injuries. However, assessing such systems through vehicle collision tests is limited, as this approach cannot cover the wide range of accident scenarios. To address this gap, identifying and generating representative pre-crash scenarios from real-world accidents provides key boundary conditions for the setup of virtual test scenarios. In this study, we used the Future Mobile Traffic Accident Scenario Study (FASS) dataset to reconstruct 112 two-wheeler accidents. For each case, we extracted pre-crash dynamic information, static attributes, and environmental context. An autoencoder was employed to encode high-dimensional features of scenarios, and K-means clustering was applied to categorize the accidents into eight representative pre-crash scenarios. For each scenario, we examined the motion states of participants and further compared the behaviors and injuries between motorcycle and bicycle accidents. The results show that motorcycles have a higher average pre-crash speed than bicycles (22.4 vs 16.2 km/h), resulting in a greater proportion of Maximum Abbreviated Injury Scale (MAIS) 4+ cases (35.6% vs 18.9%). Furthermore, a Gaussian Mixture Model was applied to fit the scenario features. This model was then used to randomly generate the initial conditions of pre-crash scenarios, including the positions, speeds, and yaw angles of vehicles and two-wheelers. The proposed scenario generation method was first applied to pre-crash scenario construction for integrated safety systems tests, and it can create diverse, statistically grounded scenarios that reflect real-world accident distributions. These scenarios can broaden the test scope of integrated safety systems and support their implementation.

Introduction

Currently, two-wheeler traffic safety still faces severe challenges. According to the World Health Organization, an estimated 1.19 million people worldwide died in road traffic accidents. Among them, powered two/three-wheelers and bicyclists accounted for 27% [1]. Data from the National Bureau of Statistics of China also show that there were a total of 75,758 traffic accidents involving two-wheeled motorcycles throughout the year, resulting in 14,836 deaths[2]. With the continuous development of integrated active and passive vehicle safety technologies, a new solution has emerged to enhance the road traffic safety of two-wheeler riders. Such systems achieve coordinated protection throughout the entire accident process by effectively integrating the perception and decision-making of active safety systems with the collision protection of passive safety systems. The systems usually intervene within a critical time window of about 2 seconds

before the collision. The active safety systems first attempt to prevent the collision from happening. When a collision is unavoidable, the systems can dynamically adjust the trigger strategy and protection parameters of passive safety devices based on real-time perception information, thereby minimizing the risk of injury to vulnerable road users (VRUs) when a collision occurs[3,4]. Therefore, for the development and testing of related systems, it is necessary to rely on vehicle-to-two-wheeler interaction scenarios with a high level of urgency. Also, it is essential to conduct in-depth research on the pre-crash scenarios corresponding to real collision accidents.

By conducting cluster analysis on real-world vehicle-two-wheeler collision scenarios, it is possible to identify high-frequency and typical collision parameters. The results can reveal the common characteristics and inherent laws of accident. Therefore, it provides the key scenario basis for the development and verification of the active and passive integrated safety system, and supports the effective application of such system in the real road environment. In the traditional scenario clustering research, the clustering variables are mostly limited to static parameters such as road type, accident time, and visibility. On the contrary, the dynamic attribute parameters of traffic participants (such as driving direction, velocity change, etc.) usually rely on manual annotation[5,6], which makes it difficult to completely capture the dynamic evolution process of accidents. With the increase of the dimension of scenario variables and the introduction of time series features, traditional clustering methods (such as K-means) face significant limitations when dealing with high-dimensional and time series dynamic data[7,8], making it difficult to achieve efficient and reasonable scenario feature extraction and cluster analysis.

Based on the identification of typical vehicle-two-wheeler pre-crash scenario types, the detailed scenario parameters extracted from real accident data can further guide the generation of pre-crash test scenarios and support the development of a high-fidelity, high-coverage test scenario database. The generation methods for safety-critical scenarios can mainly be divided into two categories: knowledge-based generation and data-driven generation methods[9]. First, knowledge-based generation usually relies on expert experience to directly define scenarios[10]. When defining pre-crash test scenarios, some studies have identified typical scenario types based on accident data[11]. However, due to the lack of in-depth analysis of the distribution characteristics of key scenario parameters, it remains difficult to achieve a quantitative definition of the test scenarios. In fact, most existing tests of vehicle passive safety systems are still limited to a few fixed operating conditions defined by regulations[12]. Secondly, data-driven generation is usually based on large scenario datasets and utilizes various machine learning methods to generate diverse test scenarios[13,14]. However, due to the scarcity of real accident data,

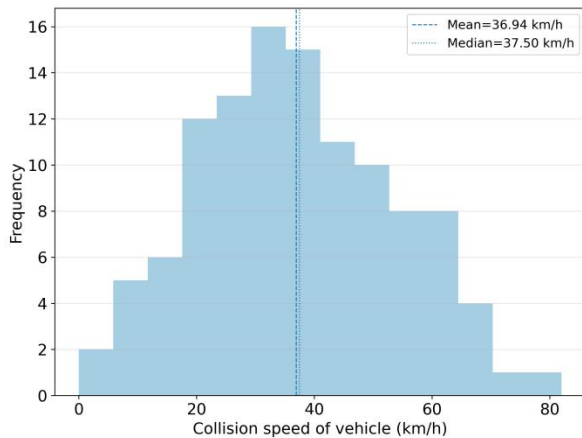
existing generative models are often not trained on real accident scenarios, resulting in generated high-risk scenarios that still lack authenticity and rationality. Therefore, in order to effectively generate pre-crash test scenarios suitable for evaluating integrated active and passive safety system, it is necessary to systematically consider the coupling relationship and joint distribution characteristics between high-dimensional scenario features based on the real accident data[15], thereby achieving more realistic and efficient scenario generation.

Methodology

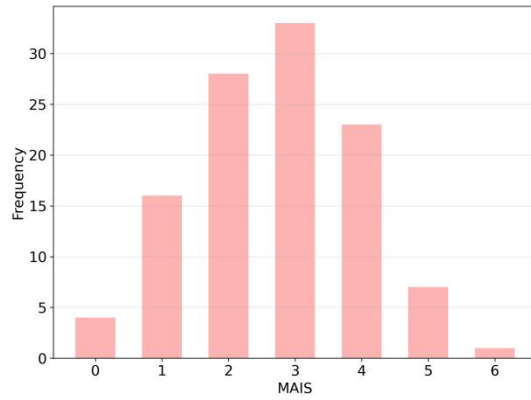
Dataset description

This study is based on the data of 112 vehicle-two-wheeler accidents in the Future Mobile Traffic Accident Scenario Study (FASS) database of the China Automotive Engineering Research Institute Co., Ltd., including 59 two-wheeled motorcycle accidents and 53 two-wheeled bicycle accidents. The relevant accident cases were reconstructed through the PC-Crash simulation software, completely recording the accident information throughout the entire process from pre-crash to post-crash. Specifically, this accident dataset covers multi-dimensional data such as accident vehicles, two-wheelers, and road environments, which provides an important basis for the study of vehicle-two-wheeler pre-crash test scenarios. The specific data content of each accident is as follows:

1. Accident vehicle information: trajectory, speed, heading angle sequence of the accident vehicle within 5 seconds before the collision, and the type of the accident vehicle. The distribution of collision speed of accident vehicles in the dataset is shown in Figure 1(a).
2. Two-wheeler information: trajectory, speed, heading angle sequence of the two-wheeler within 5 seconds before the collision, the age, gender of the two-wheeler rider, and the injury level of each part of the body (classified by the Abbreviated Injury Scale, AIS). The distribution of Maximum Abbreviated Injury Scale (MAIS) of two-wheeler riders in the dataset is shown in Figure 1(b).
3. Road environment: type of road section and signal conditions at the accident site.



(a) Collision speed distribution of accident vehicles



(b) MAIS distribution of two-wheeler riders

Figure 1. Distribution of dataset parameters

Cluster variable selection

Based on the accident dataset, this study conducted cluster analysis, aiming to determine the typical pre-crash scenarios corresponding to vehicle-two-wheeler accidents. Variables used to describe scenario characteristics can be divided into static variables and dynamic variables. Static variables refer to parameters that do not change with time in each case, such as the type of vehicle, type of road, etc. Dynamic variables refer to parameters that evolve over time, such as the trajectory sequences and speed sequences of traffic participants. In this study, static variables and dynamic variables listed in Table 1 are selected for clustering analysis. These variables represent the accident scenario information from 3 dimensions: accident vehicles, two-wheelers and the road environment. The motion attributes of traffic participants such as displacement, speed and acceleration are already implicitly expressed in their trajectory sequences, which can effectively depict the dynamic interaction characteristics between vehicles and two-wheelers before the accident occurs. Pearson correlation coefficient analysis indicates that the static variables have a relatively low correlation and can be independently used as clustering variables.

Table 1. Description of variables for clustering

Variable		Category
Dynamic variables	Trajectory sequence of the accident vehicle	The x and y coordinate sequence of the accident vehicle from 5 seconds before the collision to the moment of the collision (sampling frequency = 100 Hz)
	Trajectory sequence of the two-wheeler	The x and y coordinate sequence of the two-wheeler from 5 seconds before the collision to the moment of the collision (sampling frequency = 100 Hz)
Static variables	The relative movement direction of the accident vehicle to two-wheeler	The two-wheeler crosses from the left front of the vehicle
		The two-wheeler crosses from the right front of the vehicle
		The two-wheeler travels in the same longitudinal direction as the accident vehicle

Type of the accident vehicle	The two-wheeler travels in the opposite longitudinal direction as the accident vehicle
	Small vehicle (sedan, roadster)
	Medium vehicle (SUV)
Type of road	Large vehicle (MPV, miniVAN)
	Straight section (no intersection)
	Intersection (T-junction, intersection, multi-branching intersection, interchange)
Signal condition	Entrance and exit section (ramp entrances, real estate or road entrances and exits)
	With signal
	Without signal

Typical accident scenario clustering

Due to the high dimension of scenario variables and the presence of time-series data, such as the trajectory sequences of the accident vehicle and the two-wheeler, this study adopts a hybrid approach that combines an unsupervised machine learning model (an autoencoder) with a traditional clustering algorithm (K-means). This method enables effective extraction of temporal features from accident scenarios and facilitates the fusion of high-dimensional scenario features.

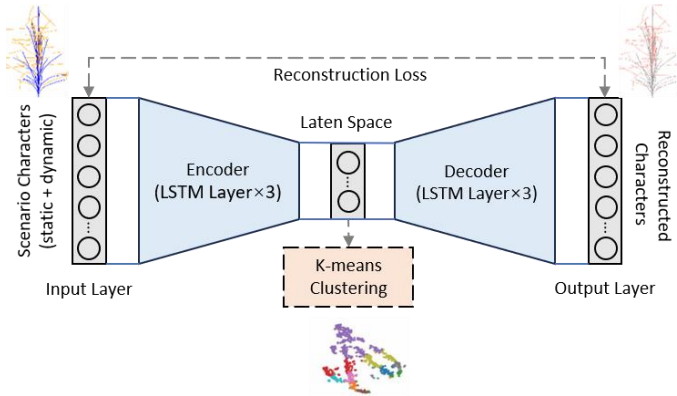


Figure 2. Structure of autoencoder network

Through the encoder-decoder structure and the stacking of its internal multi-layer long short-term memory (LSTM) network, autoencoder can realize the deep feature extraction and nonlinear dimensionality reduction of high-dimensional time series trajectory data. Its network structure is shown in Figure 2, and the detailed architecture and hyperparameter settings are provided in Table 2.

Table 2. Model architecture and hyperparameter settings

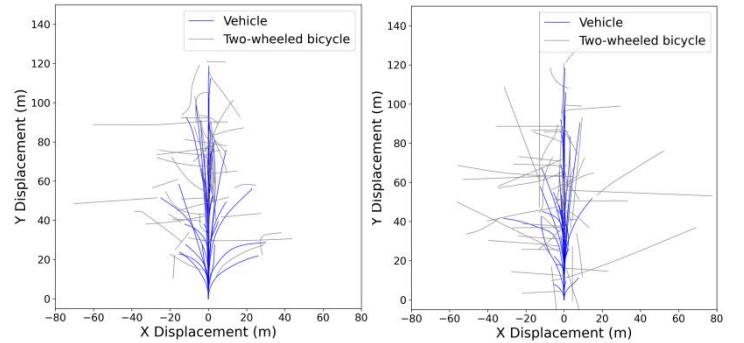
Model	Scenario Clustering		Scenario Generation	
	Encoder	Decoder	Encoder	Decoder
Layer type	LSTM	LSTM	Dense	Dense
Number of layers	3	3	2	2

Layer dimensions	[8, 8, 8]	[8, 8, 8]	[16, 8]	[8, 16]
Latent dimension	8			
Optimizer	Adam			
Batch size	16			
Learning rate	0.001			

In this network, the encoder gradually compresses the input time series data through multiple layers of LSTM units, mapping it into fixed-length embeddings in a latent space. This process effectively suppresses observation noise and redundant information while retaining key dynamic patterns. Then, the decoder performs reverse reconstruction based on the latent vector, and gradually upsamples by stacking the LSTM layer and the linear output layer to recover the reconstructed sequence consistent with the original input dimension. The entire network is trained in an unsupervised manner by minimizing the mean squared error (MSE) reconstruction loss (Equation 1). This process forces the latent variables to preserve the key structural information and temporal dependencies in the original data, thereby enabling effective compression and representation learning of high-dimensional features.

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{n=1}^N \frac{1}{T} \sum_{t=1}^T \frac{1}{F} \sum_{f=1}^F (x_{n,t,f} - \hat{x}_{n,t,f})^2 \quad (1)$$

where $x_{n,t,f}$ denotes the ground-truth (input) value of feature f at time step t for the n -th case, $\hat{x}_{n,t,f}$ denotes the corresponding reconstructed (output) value by the autoencoder, N is the number of cases, T is the number of time steps in each sequence, and F is the number of selected features.



(a) Two-wheeled bicycle accidents (b) Two-wheeled motorcycle accidents

Figure 3. Trajectory distribution of accident vehicles and two-wheelers within 5 seconds before crash (standardized)

To ensure that the autoencoder can effectively learn the temporal features contained in the trajectories of the accident vehicle and the two-wheeler and avoid introducing noise interference due to the discrete trajectory distribution among different cases, the trajectory data of the two types of traffic participants were first standardized and preprocessed. Firstly, taking 5 seconds before the crash as the starting time, the coordinates of the accident vehicle at this time are set as the origin of the coordinate system, and its heading angle is normalized to the positive direction of the Y-axis by a rotation transformation. Secondly, the trajectory of the two-wheeler in the same case is also subjected to the same coordinate transformation to maintain the relative consistency of the motion relationship. The trajectory distribution after

standardized processing is shown in Figure 3. When integrating heterogeneous feature inputs, static variables such as the type of road and the type of accident vehicle are expanded in dimension through feature replication to be consistent with the dimensions of dynamic time-series data. Subsequently, these features are concatenated at the end of the feature vector of the trajectory sequence, thereby constructing a unified high-dimensional joint feature representation, providing structured input for the subsequent multimodal fusion modeling.

After obtaining the embeddings in the latent space, in order to eliminate the possible multicollinearity among features and further improve the efficiency and robustness of the subsequent clustering process, this study further adopts principal component analysis (PCA) to perform linear dimensionality reduction and de-correlation processing on the latent features extracted by the Encoder.

Generation of typical pre-crash test scenarios

After determining the typical vehicle-two-wheeler collision accident scenarios, this study adopts an in-depth generation method based on autoencoder. By modeling and sampling the distribution of accident data in the latent space corresponding to the high-dimensional scenario parameters, it realizes the generation of various typical pre-crash test scenarios. This method can effectively capture the complex coupling relationships among high-dimensional scenario features, implicitly learn the joint probability distribution among parameters, and thereby generate test scenarios that are both realistic and diverse by sampling latent variables. The specific process of this generation method is as follows: first, the trained Encoder is used to map the standardized high-dimensional scenario parameters of real accidents into low-dimensional embeddings. In the latent space, the embeddings corresponding to each cluster of scenarios are separately fitted using a Gaussian Mixture Model (GMM) via maximum-likelihood estimation, with a full covariance structure. The number of mixture components in each GMM is determined by minimizing the Bayesian Information Criterion (BIC) for the corresponding cluster. Subsequently, latent vectors are randomly sampled from the GMM distribution of each cluster to generate samples corresponding to various typical scenarios. Then, the sampled latent vectors are fed into the Decoder, and the output results are de-normalized to reconstruct the scenario parameters with actual physical meaning.

During model training, the inputs consisted of the scenario conditions (i.e., the coordinates, speed, and heading angle of the crash-involved vehicle and the two-wheeler, as well as rider's age and gender) for each accident case, sampled at 0.1 s intervals over the 5–2 s pre-crash window. This strategy expands the dataset and ensures that the initial time step of each generated scenario falls within the 5–2 s pre-crash interval. Given that the model inputs at this stage do not incorporate temporal features, the LSTM layers in both the encoder and decoder were replaced with fully connected layers, and the MSE was also adopted as the reconstruction loss. The detailed model architecture and hyperparameter settings are provided in Table 2.

All model training and evaluation were conducted on a single desktop workstation equipped with an NVIDIA GeForce RTX 4060 Ti GPU (8 GB VRAM), an Intel Core i7-13700KF CPU (3.40 GHz), and 32 GB RAM.

Results

Clustering results

In the clustering of embeddings in the latent space, two classical clustering methods, K-means and K-medoids are compared using average silhouette coefficient (ASC) to evaluate the performance of the clustering results. It should be noted that K-means takes the virtual centroid (the mean vector of intra-cluster samples) as the cluster center, which features high computational efficiency and is suitable for large-scale continuous numerical datasets, yet it is sensitive to outliers; by contrast, K-medoids selects real samples within clusters as medoids, which has strong robustness against noise and supports diverse distance metrics for mixed-type data, but its computational complexity is relatively high and it is more applicable to small and medium-sized datasets. The ASC comprehensively reflects both the cohesion of samples within clusters and the separation between clusters, with values ranging from -1 to 1 . A higher ASC value indicates better clustering performance. Figure 4 presents the ASC values of the two methods under different numbers of clusters. Based on the comparison, the K-means algorithm is selected, and the number of clusters is set to eight.

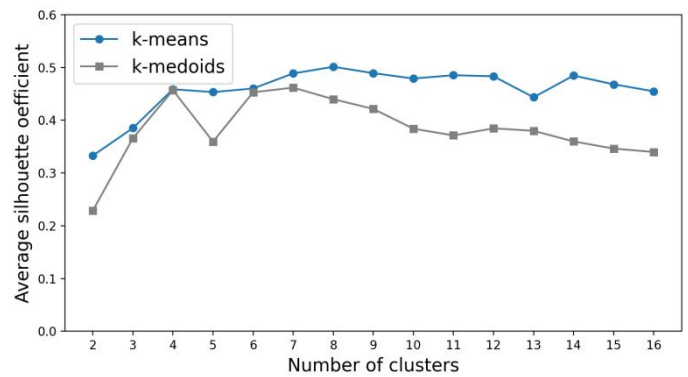


Figure 4. ASC of K-means and K-medoids by different number of clusters

Based on the final clustering results, the occurrence frequency of each variable in each cluster was statistically analyzed, and the results are shown in Table 3. The values with the highest frequency are marked in bold font, and these are the significant categories. Table 3 shows the eight typical vehicle-two-wheeler accident scenarios obtained through clustering. Among them, the driving direction of the accident vehicle and the average speed of the vehicle were not directly regarded as clustering variables, but were implicitly contained in the dynamic features of the trajectory sequence of the accident vehicle.

The specific descriptions of 8 typical vehicle-two-wheeler accident scenarios are as follows:

Scenario 1: At an intersection with a traffic signal, a small vehicle drives straight forward with medium-high speed (average speed=60.78 km/h) and collides with a two-wheeler crossing from the right.

Scenario 2: At a straight section with a signal, a small vehicle drives straight forward with medium speed (average speed=55.53 km/h) and collides with a two-wheeler crossing from the left.

Scenario 3: At a straight section with a signal, a large vehicle drives straight forward with medium speed (average speed=55.31 km/h) and collides with a two-wheeler crossing from the left.

Scenario 4: At an intersection without a signal, a small vehicle drives straight forward with medium speed (average speed=44.54 km/h) and collides with a two-wheeler in the same longitudinal direction.

Scenario 5: At a straight section with a signal, a small vehicle drives straight forward/turn left with low speed (average speed=33.23 km/h) and collides with a two-wheeler in the opposite longitudinal direction.

Scenario 6: At a straight section without a signal, a small vehicle drives straight forward with low speed (average speed=32.58 km/h) and collides with a crossing two-wheeler.

Scenario 7: At a straight section with a signal, a medium vehicle drives straight forward with low speed (average speed=26.96 km/h) and collides with a two-wheeler crossing from the left.

Scenario 8: At an entrance and exit section without a signal, a small vehicle drives straight forward with low speed (average speed=16.73 km/h) and collides with a two-wheeler crossing from the left.

Table 3. Summary of clustering results

Variable	Category	Cluster1	Cluster2	Cluster3	Cluster4	Cluster5	Cluster6	Cluster7	Cluster8	Total
Accident vehicle driving direction	Straight	15	12	13	17	8	9	13	5	92
	Turn left	2	0	0	0	6	4	2	0	14
	Turn right	0	0	0	2	1	0	1	2	6
Accident vehicle average speed	High speed	9	3	5	2	1	0	0	0	20
	Medium speed	7	9	7	10	2	3	2	0	40
	Low speed	1	0	1	7	12	10	14	7	52
The relative movement direction of the accident vehicle to two-wheeler	Cross from left	6	9	7	1	0	7	16	5	51
	Cross from right	10	3	4	2	4	6	0	2	31
	Same longitudinal direction	1	0	2	13	0	0	0	0	16
	Opposite longitudinal direction	0	0	0	3	11	0	0	0	14
Type of the accident vehicle	Small vehicle	15	10	0	13	12	10	6	5	71
	Medium vehicle	2	1	5	6	3	3	8	2	30
	Large vehicle	0	1	8	0	0	0	2	0	11
Type of road	Straight section	4	11	9	3	12	13	16	1	69
	Intersection	10	0	2	12	3	0	0	1	28
	Entrance and exit section	3	1	2	4	0	0	0	5	15
Signal condition	With signal	11	11	13	5	14	0	12	0	66
	Without signal	6	1	0	14	1	13	4	7	46
Total		17	12	13	19	15	13	16	7	112

Note: A vehicle with an average speed greater than 60 km/h within 5 seconds before a crash is defined as high speed, within the range of 40 to 60 km/h as medium speed, and less than 40 km/h as low speed.

This study further clustered the accident cases of two-wheeled bicycles and motorcycles separately to determine the differences in vehicle-two-wheeler interaction patterns and scenario parameters in the two types of accidents. Table 4 shows the distribution of vehicle-two-wheeler interaction modes in two types of accidents. In general, the distribution of interaction modes between vehicles and two-wheelers was consistent in the two types of accidents. Specifically, in two-wheeled motorcycle accidents, the proportion of accidents caused by crossing was higher than that caused by two-wheeled bicycles. Two-wheeled motorcycles also had a higher proportion of accidents than two-wheeled bicycles when driving in the opposite direction, while two-wheeled bicycles had a higher proportion of accidents when driving in the same direction as the vehicle. The reason for the above results may be that the speed of motorcycles is usually significantly higher than that of bicycles. In scenarios involving path conflicts such as crossing and driving in the opposite direction with the vehicle, collisions are more likely to occur

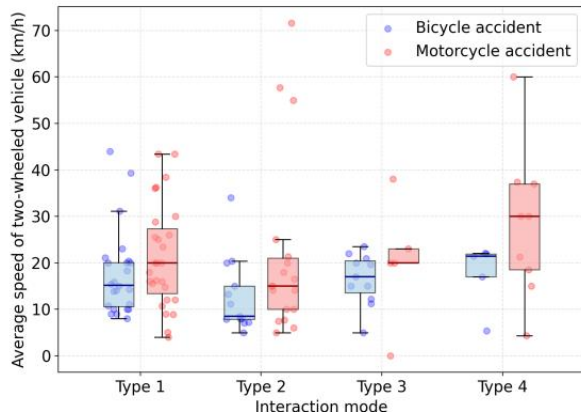
due to the greater relative speed, which requires a longer braking distance and longer reaction time to avoid collision.

From the perspective of the relative movement direction of the accident vehicle and the two-wheeler, most accidents occur when the two-wheeler crosses the road, while the number of accident cases is relatively low when the vehicle is driving in the same direction or opposite direction. Furthermore, the data indicate that the number of collisions involving the accident vehicle when driving straight is far greater than that when turning left or right. It is worth noting that the distribution of accidents in left-turn and right-turn situations is not symmetrical. The number of collisions caused by vehicles turning left is significantly higher than that when turning right. The reason may be that, when turning left, vehicles need to cross the opposite lane and intersect with multiple traffic directions at the same time. Compared with turning right, drivers face more complicated decisions and higher accident risks.

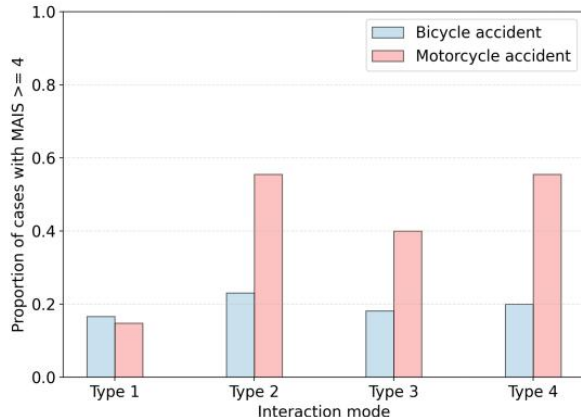
Table 4. The distribution of vehicle to two-wheeler interaction modes

		Accident vehicle driving direction			Total
		Accident vehicle driving straight	Accident vehicle turning left	Accident vehicle turning right	
The relative movement direction of the accident vehicle to two-wheeler	Cross from left (Type 1)	33.7% (40.7%)	5.7% (3.4%)	5.7% (1.7%)	45.1% (45.8%)
	Cross from right (Type 2)	20.8% (25.4%)	3.8% (5.1%)	0.0% (0.0%)	24.6% (30.5%)
	Same longitudinal direction (Type 3)	18.9% (8.4%)	0.0% (0.0%)	1.9% (0.0%)	20.8% (8.4%)
	Opposite longitudinal direction (Type 4)	5.7% (10.2%)	3.8% (3.4%)	0.0% (1.7%)	9.5% (15.3%)
Total		79.1% (84.7%)	13.3% (11.9%)	7.6% (3.4%)	100.0% (100.0%)

Note: the numbers outside the brackets represent two-wheeled bicycles; The numbers in brackets represent two-wheeled motorcycles



(a) Comparison of two-wheeler average speed



(b) Comparison of max injury levels (MAIS)

Figure 5. Comparison of two-wheeler bicycle and motorcycle accidents

The four relative movement directions of the accident vehicle to two-wheeler, namely, crossing from left, crossing from right, traveling in the same longitudinal direction, and traveling in the opposite longitudinal direction, are regarded as four types of vehicle-two-wheeler interaction modes and are numbered 1 to 4. Figure 5 compares the average speed of two-wheelers and the MAIS distribution of riders in two-wheeler bicycle accidents and two-wheeler motorcycle accidents among these four types of interaction modes. As shown in Figure 5(a), in the four interaction modes, the average speed of the two-wheeled motorcycle in the 5 seconds before the crash is higher than that of the two-wheeled bicycle. This result is consistent with the conclusion of the aforementioned accident cause analysis. Overall, the average speed of two-wheeled motorcycles in all interaction modes within 5 seconds

before a crash was 22.4 km/h, while that of two-wheeled bicycles was 16.2 km/h. Figure 5(b) shows that, except for the scenario in which the two-wheeler crosses from the left, the proportion of motorcycle riders sustaining severe or fatal injuries (MAIS 4+) is significantly higher than that of bicycle riders in the other three interaction modes. Among all interaction modes, the proportion of MAIS 4+ injury cases among two-wheeled motorcycle riders is 35.6%, while the corresponding proportion of cases among two-wheeled bicycle riders is 18.9%.

Injury analysis based on scenarios

In this study, ordinal logistic regression was used to analyze the correlations between parameters such as collision conditions (collision speed of the accident vehicle, collision speed of the two-wheeler, and collision offset), vehicle attributes (type), and rider attributes (age, gender) and the MAIS of riders after the collision. The relevant analysis results are shown in Table 5.

Table 5. Ordinal logistic regression results for MAIS of riders

Variable	Coef	Std.Err	Odds ratio	p-value
Collision speed of vehicle	0.550	0.197	1.733	0.005 (**)
Collision speed of two-wheel vehicle	0.201	0.183	1.223	0.272
Collision offset	0.359	0.184	1.432	0.051 (*)
Small vehicle (relative to the base group: large vehicle)	-0.864	0.616	0.421	0.161
Medium vehicle (relative to the base group: large vehicle)	-0.825	0.653	0.438	0.207
Age of rider	0.372	0.191	1.450	0.052 (*)
Gender of rider (relative to female)	0.167	0.411	1.181	0.685

Note: Collision offset refers to the distance between the initial collision position and the vehicle's central axis. In this table, all continuous variables are standardized to distributions with a mean of 0 and a standard deviation of 1.

The ordinal logistic regression results showed that the collision speed of the accident vehicle was significantly positively correlated with the injury level of the rider ($p < 0.05$). Under the premise of the standardization of continuous variables, the ratio of the occurrence of higher injury level increases about 1.73 times for each standard deviation increase in the speed of the collision vehicle. In addition, the collision offset ($p = 0.051$) and the rider's age ($p = 0.052$) were at the marginal significant level ($p \approx 0.05$), suggesting that a more deviated collision position from the vehicle's central axis and an older age may be associated with more severe injury.

The accident cases were classified into three groups according to the age of the riders: young (age ≤ 30), middle-age ($30 < \text{age} \leq 60$), and old (age > 60). Figure 6 shows the MAIS distribution of riders in the three age groups under different vehicle collision speeds and different offsets. As shown in Figure 6, under the conditions of higher collision speed and offset, the injury level of riders generally shows an increasing trend. Meanwhile, as the age of riders increases, the proportion of cases corresponding to high injury levels also gradually rises. It is worth noting that some cases still deviate from the above collision condition - injury relationship, indicating that the severity of injury may also be affected by whether riders wear protective helmets, the head contact point after collision, the vehicle's structure design, and other factors. In addition, among the four types of interaction modes, the proportion of severe and fatal injuries to the riders (MAIS 4+) is the highest when the two-wheeler is traveling in the opposite longitudinal direction to the accident vehicle. The reason is that the relative speed between the two participants is usually greater in this interaction mode.

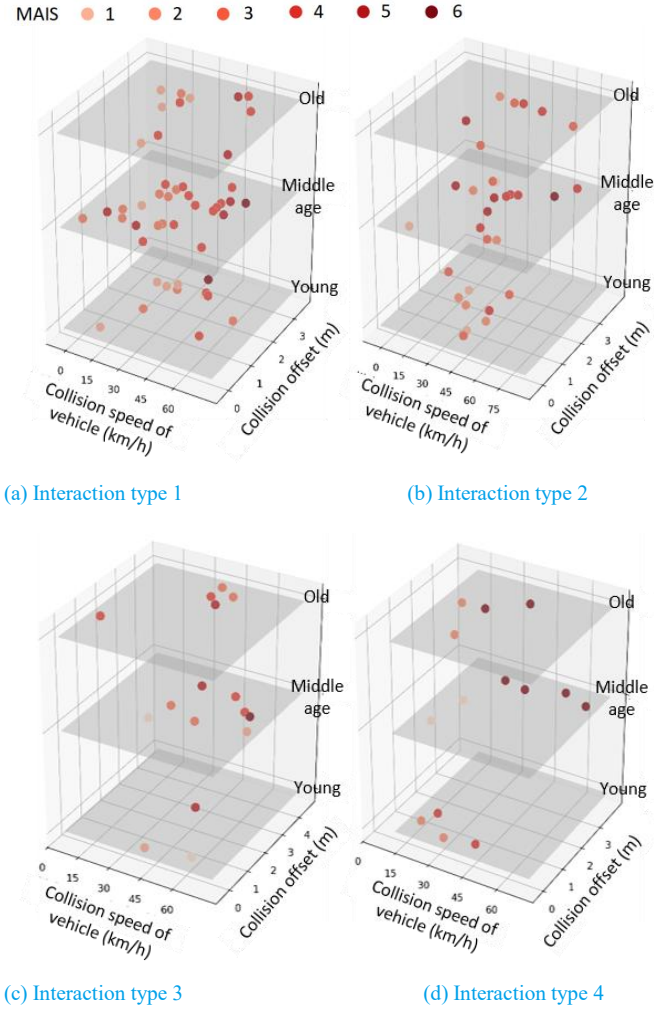


Figure 6. The distribution of injury level of riders

Generation results of typical pre-crash test scenarios

Based on the method of random sampling from each cluster in the latent space, this study has achieved the generation of 8 typical vehicle-two-wheeler pre-crash scenarios shown in Table 3. To quantitatively validate the realism of the generated scenarios, we compute the Fréchet Distance (FID) in the latent space between real accident samples and

generated samples for each cluster (Equation 2). Additionally, we introduce a weighted Kullback-Leibler (KL) divergence to characterize the overall similarity of scenario conditions (i.e., the coordinates, speed, and heading angle of the crash-involved vehicle and the two-wheeler, as well as rider's age and gender) between the generated and real-world pre-crash scenarios (Equation 3). The detailed validation results are provided in Table 6.

$$D_{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr} \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{\frac{1}{2}} \right) \quad (2)$$

$$D_{KL} = \sum_{n=1}^N \frac{1}{N} \int p_n(u_n) \log \frac{p_n(u_n)}{q_n(u_n)} du \quad (3)$$

where μ_r (μ_g) and Σ_r (Σ_g) are the mean vector and covariance matrix of the latent samples z drawn from $p(z)$ ($q(z)$), $p(z)$ and $q(z)$ are the latent space distributions of the real accident samples and the generated samples, respectively, $\|\cdot\|_2$ denotes the Euclidean norm, and $\text{Tr}(\cdot)$ denotes the trace operator of matrix; $p_n(u_n)$ and $q_n(u_n)$ denote the distributions of the n -th scenario condition feature u_n in the real and generated samples, respectively, and N is the number of scenario condition features used for scenario generation. In general, lower D_{FID} and D_{KL} indicate greater similarity between the distributions, with a value of 0 implying that the two distributions are identical.

Figure 7 illustrates four types of generated scenarios to demonstrate the effectiveness of the proposed scenario generation model. Specifically, the four types represent the following interaction modes: (a) the two-wheeler crossing from the left, (b) the two-wheeler crossing from the right, (c) the two-wheeler traveling in the same direction as the vehicle, and (d) the two-wheeler traveling in the opposite direction to the vehicle, which correspond to Scenario 2, 1, 4, and 5 in Table 3, respectively. For each scenario type, three examples are provided to show the generated pre-crash initial conditions, including the relative position between the vehicle and the two-wheeler, as well as their respective speeds and heading angles. The arrows denote the initial heading directions, and the adjacent numerical annotations indicate the corresponding initial speeds. For virtual testing of vehicle-integrated safety systems involving two-wheeler riders, the initial states of the test vehicle and the two-wheeler can be configured according to these generated initial conditions. Other test configurations (e.g., road type and the presence or absence of traffic signals) can be specified based on the static scenario characteristics associated with each scenario type, as summarized in Table 3.

Table 6. Validation results of the generated scenarios

Variable	Number of GMM components	D_{FID}	D_{KL}
Scenario 1	8	0.023	0.241
Scenario 2	10	0.028	0.178
Scenario 3	10	0.030	0.281
Scenario 4	8	0.024	0.239
Scenario 5	12	0.018	0.249
Scenario 6	8	0.013	0.155
Scenario 7	10	0.031	0.295
Scenario 8	10	0.015	0.329

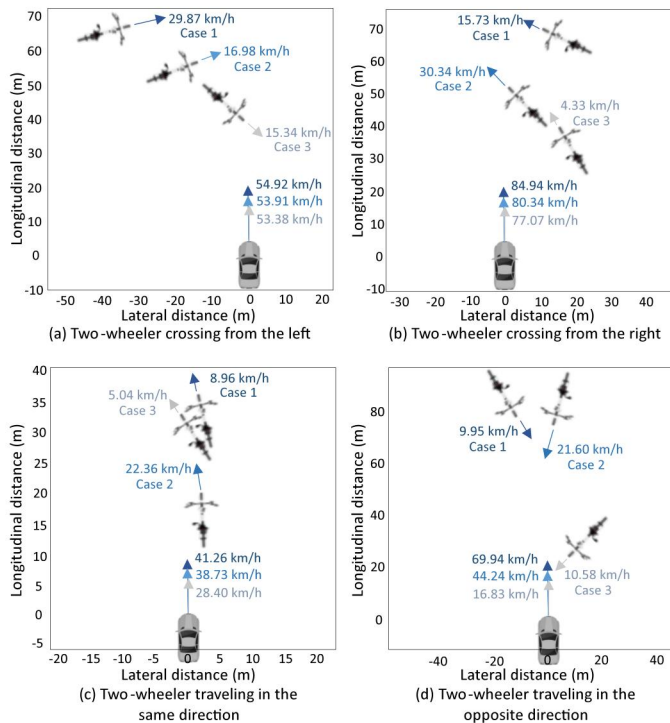


Figure 7. Example of generating initial testing conditions for a typical pre-crash test scenario

Conclusions

Based on 112 two-wheeler accident cases filtered and reconstructed from the FASS database, this study determined 8 typical vehicle-two-wheeler pre-crash scenarios by combining an autoencoder with clustering algorithms, and achieved the generation of various typical pre-crash test scenarios using the in-depth generation method. Compared with traditional accident scenario clustering methods that only consider static features of the scenarios, this study further incorporates dynamic features such as the trajectory sequences of accident vehicles and two-wheelers into the scenario clustering. It is worth noting that although features such as vehicle speed and driving direction are not explicitly used as clustering variables, their differences are implicitly characterized through trajectory sequences and naturally presented in the clustering results. This method avoids the manual extraction of accident scenario features and more effectively realizes the coupling between scenario temporal features and high-dimensional scenario parameters.

The analysis of motorcycle and bicycle crashes reveals systematic differences in both interaction modes and injury outcomes between the two types of crashes. In motorcycle accidents, the proportion of crossing and driving in the opposite longitudinal direction is higher, while in two-wheeled bicycle accidents, the proportion of driving in the same longitudinal direction is higher. Because the average speed of motorcycles is significantly higher than that of bicycles, this leads to a relatively higher severity of injuries for their riders, and a relatively larger proportion of severe and fatal injuries. These findings suggest that when developing active and passive safety systems for two-wheelers, it is necessary to distinguish the different dynamic characteristics and typical conflict modes of the two types of two-wheelers and optimize them for specific test scenarios.

The results of injury analysis showed that the accident vehicle speed was the most significant factor affecting the injury severity of riders, and it was positively correlated with the injury level, which was consistent with previous studies. The collision offset and the age are also marginally significant to riders' injury, indicating that both the collision position and the rider's physical condition have an impact on the injury outcome. There are some cases in the injury results that deviate from the overall distribution, indicating that the injury mechanism is also affected by other factors (such as whether the rider is wearing a protective helmet, the position of head contact point after the collision, the design of the structure of the vehicle, etc.). Corresponding variables should be further introduced for analysis in subsequent studies.

In addition, the pre-crash test scenario generation method based on real collision accident data proposed in this study fully considers the coupling relationships of high-dimensional scenario parameters, providing an abundant, realistic and typical virtual scenario foundation for testing of the integrated active and passive safety systems, and supporting more targeted safety performance verification and development.

In the scenario generation pipeline (encoder embedding—GMM sampling—decoder reconstruction), limitations in representation learning or imperfect embedding fitting can affect the realism and diversity of the generated scenarios. In particular, the number of GMM components critically influences the trade-off between diversity and generalization (underfitting with too few components versus overfitting with too many). To improve robustness, we standardized trajectories, de-correlated latent features using PCA, and expanded the training set by sampling the 5–2 s pre-crash window at 0.1 s intervals.

This study still has some limitations. Due to the limited amount of data, the number of cases in some subcategory scenarios obtained through clustering is relatively small, which may lead to limited representativeness of the clustering results in some long-tail traffic scenarios in the real world. Additionally, since the proposed framework is designed to generate scenarios constrained to the empirical distribution of the observed accident dataset, it does not specifically generate out-of-distribution scenarios that are not represented in the original data. In the future, it can be considered to further expand the data quantity through multi-source data fusion. For the generation of typical pre-crash test scenarios, future research can enable efficient synthesis of traffic participants' trajectories and road topological structures through data augmentation and more advanced conditional generative models (e.g., conditional variational autoencoders or diffusion-based models), while ensuring more robust and physically plausible generation of out-of-distribution test scenarios. These advances may provide more abundant scenarios for the test of the active and passive integrated safety system of intelligent vehicles.

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Contact Information

Guojie Wang, Jiaxin Liu and Liangliang Shi are with the State Key Laboratory of Intelligent Vehicle Safety Technology and China Automotive Engineering Research Institute Co. Ltd, Chongqing, 401122, China. (e-mail: wangguojie@caeri.com.cn; liujiaxin01@caeri.com.cn; ll.shi@outlook.com)

Xin Gao, Siyuan Liu, Quan Li, and Bingbing Nie are with the School of Vehicle and Mobility, Tsinghua University, Beijing 100084, China. (e-mail: gaox24@mails.tsinghua.edu.cn; siyuan-122@mails.tsinghua.edu.cn; lq1911@mails.tsinghua.edu.cn; nbb@tsinghua.edu.cn).

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